A CONCEPT NOTE

1 NAME OF THE ORGANIZATION

Turning Drylands Green FOUNDATION

2 TITLE OF THE PROPOSED PROJECT

An Initiative " to Rehabilitate Dryland Catchments towards the Restoration of Denuded Landcapes"

3 POTENTIAL COLLABRATORS

Among the Stakeholders to be approached include: -

3.1-GoK (Treasury, Environment, ASALs, Agric, Liv &Water),
3.2-Council of Governors
3.3-International Union of Conservation of Nature (IUCN)
3.4-Japanese Embassy
3.5-Habitat, UNDP, UNEP, FAO
3.6-Conservation International
3.7-OXFAM GB
3.8-Peace Parks Foundation, et al.
3.9- African Forest Landscape Restoration Initiative
3.10-World Resource International (WRI),
3.11-Imara Intergrated Farmers' Solutions,
3.1.2-Safaricom, *et al*.

4 CONTEXT

4.1- Oxfam, in its report, revealed that millions of individuals in Africa are grabbling with substantial losses, approximately KES 11.2 billion, solely, from the loss of their livestock. Yet, grasslands, savannahs and shrub lands are a lifeline for millions of pastoralists and, indirectly, support more than 200 million people. Millions of hectares of these ecosystems have deteriorated, jeopardizing water sources, carbon sequestration as well as community livelihoods (*Wambui Mbuthia, SN November 19, 2023*). Moreover, it is an undisputable fact that degradation of these Catchment areas is the largest contributor to conflicts over pastures and water often experienced.

4.2-In Kenya, with an area of about 587,000 Sq. Km, *84% is categorized as Arid and Semi-Arid Lands* (*ASALs, reffered to as Drylands*). This 84% is characterized by ***harsh living conditions and a fragile eco-system,*** which experiences extremities of weather, with sporadic rainfall that drains away. *This area is, ideally, suited for pastoralism; however, over time, some agro-pastoralism has been going on through extensive cultivation, but accompanied by crop failure. This has led to more soil destruction through fragmentation, and use of unsuitable water for irrigation*.

Occasionally, this vast region receives heavy downpours, with subsequent flash floods causing further damage, ***in terms of massive soil erosion, structural destruction, as well as loss of human & livestock life.***
Overtime, Kenyan ASALs have been left denuded because of climate change, as a result of climate variability; a global phenomenon.

4.3-*Even the 'safe 16%', like the Maasai Mara Catchment*[1]*, has faced such "unfamiliar devastation", as recently as 2019* - when it was hit by drought, due to climate variability and lack of eco-system sustainability. It seems the effects are spreading across, and the Country is already, increasingly, get afflicted by disasters, which pose challenges to any development.

[1] ***the water collection, especially rainwater, over a natural drainage waterway.***

4.4-But, the Country can ill afford to treat this 84% as just another statistic, due to (a) its enormous contribution to National Food Security, (b) 10% of the National GDP, while accounting for about 40% of the agriculture sector's portion, (c) 25% of the populace it supports, (d) employs >50% - both directly and indirectly of the Country's labor force.
The sector also accounts for >10% of the country's export earnings. But, most of all, these ASALs supply the bulk of the beef consumed in Kenya.

4.5-The contribution of the livestock sector to the Kenyan economy was valued at KES 188.2 billion, out of the Agriculture Sector's 961.3 (*KNBS Reports, 2022*). *Nevertheless, the true value of this sub-sector's contribution is higher, if unrecorded slaughter figures, home consumption, hides/skins, as well as dairy/dairy products, are taken into account.*

4.6-And, The African Climate Summit-2023 (***ACS23***) in Nairobi couldn't have summarized it better, than *"Africa contributes the most yet losses most (US 120B annually) in costs of mitigating the impacts of Climate Variabilities"……*

4.7-**The Nexus Between Water, Environment & Livelihoods:** *The recurying drought in the ASALs, with reactive supply of relief, water & or accompanied with livestock offtake has not, will not address drought that seem to be expanding, without firstly addressing how we could avail water to turn around environment in these areas. Theres a definite need to address this challenge outside what has and continue to be done in the context of* ***harvesting the enormous amounts of water that ocassionally pour (but drains off) in these areas****; TDG has the technical knowhow and the capacity.*

5 RATIONALE *(why)*

5.1-Thus, drought, floods & famine in Kenya, are not new; while rainfall patterns used to be predictable, *with a degree of certainty*; not anymore. Even the Ministry of Agriculture used to Gazette times of planting, and due to this uncertainty, *this advisory was abandoned*. What is, therefore, *'new' is its decreasing period of frequency & repeatability, as well as severity; initially recurring in more than 10 - 15 years before the 1960's, to currently, less than two years!* During these periods, the Government is compelled to declare National Disasters that affect an ever increasing population. *The frequency/repeatability/severity don't even allow recovery from one disaster to another, thus cumulatively aggravating the problem.*

Incidentally, after the rains have stopped, with the ***swollen Waterways and Rivers receded, 'it's as if the climate-driven disaster never happened…'*** *However, for many, the crisis goes on as they struggle to rebuild their lives.* ***In June 2024, scientists from World Weather Attribution stated, "climate change contributed to the disastrous heavy downpours, and warned that such events would become more frequent if nothing is done…."****; with this in mind, TDGI, therefore, is offering that something.*

5.2- THE PROBLEM STATEMENT: (*Landscape restoration being a national priority in Kenya that* ***is driven by an ambitious goal to restore 10.6 million hectares of degraded ecosystems by 2032)*** **"**How then **"do we Restore this** (vast, 84%) **Degraded & Denuded Landscape without, first, having to harvest rain water"** *(to enable the planting/growing those trees)***?** *Indeed, these efforts—which include growing 15 billion trees and reclaiming vital catchment areas—****aim to combat climate change, protect biodiversity, and secure community livelihoods***

5.2.1-For how long shall the Government distribute relief food, fund cattle offtake etc., whose budgets are ballooning every time…….?

5.2.2– *(In any case)* exactly how much is this limited budget spent; on how many people……?

5.2.3–Have the funds been cost-effective? How about coverage, satisfaction by the beneficiaries; has it been documented, to suggest it is/has been worthy!

5.2.4-Where is this budget anyway (*because usually, the Government compels itself to divert the limited budgetary allocations, meant for other essential operations, just to intervene in these disasters*)?

5.2.5 -Consequently, as a Country, we may never progress, as we seem to be spending colossal sums of our limited resources, 'ritually'!

5.2.6–Moreover, no one doubts the fact that there is *human dignity* when a family, or, a person, feeds themselves, & at their own time, as opposed to "*being given relief food (maize & beans*)"!

5.2.7-The interventions, seemingly, became 'knee-jack responses', as opposed to addressing the underlying cause……

5.3 – It is time "we" tackle this scourge head-on, and strategically.

5.4 – Indeed, this Initiative takes cognizance of (i) individual efforts undertaken, at whatever level, (a) towards reduction of desertification by Ms. Patricia Mumbua Kombo - with her *Par Tree Initiative*, and (b) *Green Belt Movement*, by late Prof. Wangare Mathai - *which encourages women to work together to grow seedlings, and plant trees to bind the soil, store rainwater, provide food and firewood, and receive a small monetary token for their work,* leading to their UN Recognition separately,

(ii)The Lare Catchment case in Njoro, by (late) Prof. Wanjama - under KARI, *that turned the area from relief dependency, to a net exporter of fresh vegetables/plantain bananas'*, to the Nakuru Market, and currently (iii) An effort, through communities in Turkana, on production of grass fodder for their livestock; *are vivid examples of what can, should be replicated and up scaled.*

5.5–To mitigate the impacts of drought, in 2011, the Government announced a budget of about **KES 192 Million**-*to cater for the cost of livestock interventions alone*, and at one session, because in a given year, it may be repeated (*The Standard, Mar, 2011*); ***this Initiative would hardly require such an amount per site, neither would it be repeatedly.***

However, according to research by ***Systems Change Lab***[2], roughly $ 5 trillion of climate finance/year by 2030 is necessary to prevent the worst impacts of climate change. But, current *investments & financial flows remain off-track from those levels needed to protect people, nature & biodiversity – and "we are running out of time to correct the course…" (WRI).*

5.6- While restoration of the ASALs is key, via ultimately, the planting & growing of vegetation (trees) , "*the pioneering study of progress shows how the last 25 years of accelerated reforestation around the world might significantly pay off in the second half of the 21st century.*

Using a variety of calculative methods and estimations based on satellite and temperature data from weather stations, the authors determined that forests in the eastern United States cool the land surface by 1.8 – 3.6°F annually compared to nearby grasslands and croplands, with the strongest effect seen in summer, when cooling amounts to 3.6 – 9°F. The younger the forest, the more this cooling effect was detected, with forest trees between 20 and 40 years old offering the coolest temperatures underneath.

"The reforestation has been remarkable and we have shown this has translated into lowering of the surrounding air temperature," Mallory Barnes, an environmental scientist at Indiana University who led the research, told The Guardian".

[2] ***Systems Change Lab*** *= a WRI initiative that "tracks global progress across systems and highlights current action against climate biodiversity and equity targets"; '**SCL website'** is an open data platform that will track progress on multiple indicators across a range of systems that are fundamental to meeting global climate, biodiversity and sustainability targets'.*

6 PROJECT GOALS/OBJECTIVES (*summarized logframe*)

Overall Objective	Specific Objectives	Activities	Outcomes
To rehabilitate and restore ASAL Catchments	1) To improve water availability by targeting 2 catchments through development of at least one water pan, five cut-off drains, five shallow wells etc, per catchment	a) Identify and map 2 Catchments (sites)in 7 Counties;	▪ Cartographic map of the catchment with a legend
		b) Identify 25 local labourers per site;	▪ contractor/hydrologist with a suitable profile identified
		c) Design 7 cut-off drains, 7 shallow wells, & other reservoirs;	▪ A drawing design of drainage canals, cut-off drains, water pan, or Dams,shallow wells; ▪ Reduced distances to water sources for women and school-going-children
		d) Undertake the construction water harvesting structures & harvest the rainwater	▪ Cut-off drains/shallow wells/water pans constructed to design specs
			▪ Increased availability of water for both human & livestock use
			▪ Incidences of water related diseases will have been reduced;
			▪ Improved school attendance and cognitive performances
			▪ Improved quality/quantity of water, thus reducing conflict related incidences
	2) To rehabilitate/conserve catchments	a) To rehabilitate dilapidated waterways	▪ Rehabilitated rainwater ways and checked soil erosion
		b) To plant suitable vegetation along embankments	▪ Conserved catchments to control siltation of water structures
	3) To increase overall forest cover in the catchment by 15%	a) Determine the current forest cover for the 14 Catchments;	▪ A report showing % forest cover of the catchment; ▪ Carbon sequestration created, as an extra source of income
		b) Identify suitable agro-forestry tree/fodder shrub for each Catchment	▪ A report with suitable tree/fodder tree/shrubs species ▪ Reduced destruction of catchments/soil erosion
		c) Establish 2 nurseries per catchment & enable communites to mgt;	▪ Nurseries established
			▪ Fodder tress, other trees/shrubs grown
		d) Enable communities plant & grow the respective vegetation;	▪ Reduced destruction of forest
			▪ ***Improved % of forest cover & carbon trade openedup in the ASALs***
			▪ Increased biomass of wood availability
	4) To enhance capacity of local communities in catchment restoration and management	a) To capacity build local communities in dryland forest management	▪ Improved adaptation of dryland forestry knowledge ▪ Reduced destruction of forest ▪ Catchments protected
	5) To improve/increase forage for livestock/wildlife by 100 acres per catchment	a) Determine the current pasture acreage	▪ Current pasture acreage established
		b) Identify suitable grass species for each catchment	▪ Type of grass suitable for area documented
			▪ Increase in number of healthy livestock
		c) Establish a grass pasture	▪ Pasture acreage improved
			▪ Volume of livestock products increased, hence income generated as well as food & nutrition security attained;
			▪ food availability improved
	(5) Improve modern farming techniques	a) Identify farms for piloting b) Practice some model irrigation farms	▪ Modern farming techniques adopted

7 The STRATEGY

While this approach is multi-sectorial*, implementation shall be synched; the reason it *should neither be left to ASALs (where its mandate is located), ***Water, Agriculture, Environment nor Forestry Department solely (due to 'sibling rivalry', or competing nature of activities*!). Among the key activities to be undertaken are,**

7.2.1-Completion of desk work and sharing with the respective County Government, & potential Collaborators, and if approved,

7.2.2– Benchmarking snapshot visits to Lare, in Njoro, and Ms. Gladys Mumbo farm in Mutomo, Kitui (*the initiative would gladly invite interested stakeholders to visit any of these, or both*);

7.2.3- Courtesy call on respective local administration (Internal Security & County),

7.2.4- Reconnaissance visits to identify catchments for target, survey & mapping,

7.2.5- Community Capacitation & election of officials for Catchment User Groups',

7.2.6- Excavation/Rehabilitation of rainwater drainages,

7.2.7- Excavation of cut-off drains, and holding structures (*trenches, narrow base channels, hillside ditches, drainage canals, small basins etc.),* to help contain speed/damage, help in the water harvesting, and lining them to reduce seepage,

7.2.8- Identification, bulking, planting & growing of suitable perennial *shrubs/trees, like Acacia tottollis spp,*

7.2.9- Excavation of water pans, wells or boreholes, where there's none, to ascertain water supply for irrigating the planted forage shrubs/trees, or as the case may apply, and,

7.2.10- ***Replication and upscaling of the Initiative to the rest of the drylands, as funds are availed.***

8 EXPECTED OUTCOMES:

Turning Drylands Green envisages long-term tangible outcomes that will allow recovery & opening up the ASALs for development; key among them are: -

Expected Outcomes/UN-SDGs

			SDG
1	mobilization/empowerment of communities	a)	***Communities participation will be harnessed, thus ensuring its sustainability & equity.***	5/10
2	rehabilitation of rainwater ways	a)	Conserved catchments will be achieved, thus reducing destruction by soil erosion-hence reduced siltation of water bodies & structures.	9/14
3	excavation of water harvesting structures	a)	**A significant portion of the rain water lost, will be harvested & stored** *(It was established that 'improving rainwater use efficiency can alter climate vagaries in the ASALs……' (Duba Golicha, Land Health Expert, a Standard Newspaper Publication).*	6 16
		b)	Watering points for human/livestock, forest cover & pasture acreage significantly increased to ascertain good health of livestock.	
4	harvesting the rainwater	a)	*Sustainable agricultural activities resumed, achieving food and nutrition security, thus restoring livelihoods, hence their dignity.*	2/3
		a)	**The government would have saved the billions of budgetary money, for required developmental agenda.**	17
		a)	***Schools' attendance Improved, as well as cognitive performances.***	4
		a)	**Water in the parks will be availed for wild animals; to restore tourism.**	15
5	growing of vegetation (trees, shrubs, pastures etc), making it sustainable	a)	*Incomes will be generated, hence restoring livelihoods;*	1/12
		b)	***Opening up of carbon credit trade (sequestration) in ASALs;***	
		a)	**Environment will be restored, *reduce carbon emissions by 30%.***	13
		a)	The ***new carbon trade*** shall also enhance income for pastoralists…	8

9 INNOVATIONS (*how different from previous attempts*)

9.1-This Initiative is not aware of any similar attempt, other than (i) the reccurent programs by various government departments, & (ii) the (seemingly) ***recurying knee-jack responses*** {*a) relief food supplies, b) cattle off-take, c) distribution of water (in tanks), d) human resettlements, e) cash transfers etc.*}; and, in spite of this, ***the drought, famine, floods*** – *accompanied by* ***soil erosion/landslides***, **don't seem to ebb at all**!

9.1.1-However, this Initiative took coignisance of *three previous projects*; a) NSWCP, in the early 1980's, and as the name suggests, largely involved itself with soil & water conservation structures, and was domiciled at the Ministry of Agriculture. It had a lifespan. *One wishes it was entrenched for sustainability;*

9.1.2-two recently implemented through the Ministry of Livestock, namely b) *Regional Pastoral Livelihoods Resilience Project* (RPLRP-Kenya), covering *Garissa, Isiolo, Mandera, Marsabit, Samburu, Wajir, W/Pokot, T/River, Baringo, Narok, Kajiado, Laikipia & Lamu*. It was twofold, (i) construction of water infrastructure, and (ii) rehabilitation of degraded rangelands, and, c) *Drought Resilience & Sustainable Livelihood Programme'*, (DRSLP), covering *Samburu, Isiolo, Marsabit, Baringo, West Pokot & Turkana*. It was rolled out in July 2013, and focused on improving resilience to drought by communities in the ASALs.

9.1.3-TDG may not be able to appraise *impacts of past or ongoing approaches, but what is for sure is that ravages of drought haven't shown any signs of relenting. Instead, what is read, seen and heard is the severity of drought and its repeatability, as well as the population affected; getting worse every successive episode, hence the drive for this approach. While the Ministry in-charge of ASALs reported it had spent KES 1.2 Billion on relief, in March 2023 alone* (D/N Friday, April 14, 2023*)*, this Initiative, while focusing on harvesting of rain water & restoring vegetation cover, would hardly spend such an amount to rehabilitate catchments in, at least, 7 counties, ***once and for all***…….

9.2-*This Initiative, also, intends to* ***augment*** *the Government's 30-30 Strategy, which targets to plant 15 billion trees, through the Ministry of Environment, whose objective is to* ***mitigate the effects of climate change****; but, what disturbs is when its reported "the so-n-so's in this Country have gone to plant trees in Ngong, Mau,Kaptagat Forests etc, not mentioning the huge budgets spent ……."*

10 ORGANIZATIONAL BACKGROUND

The idea behind **Turning Drylands Green**, cropped up in early 2002/3, during household needs surveys - under the auspices of the Kenya Food Security Meeting (KFSM), *among others:-*

10.1– (a) In Narok, carcasses of animals were encountered, belonging to pastoralist who had lost >95% of the >650 herd of cattle, to severe drought. He was devastated;

(b) In Samburu, a mother of about 5 children, *was trying to excavate a borehole; in the process, some goats were also jumped* into the hole to access the same trickling water, while

(c) in Baringo, *a mother was 'trying to boil wild fruit seeds* ('*just to give an impression to her children she was cooking some food'); in the process, the children got asleep….* '! These, among others, were pathetic scenes for any rational observer to behold!

10.2-This idea, only crystalized, after experiences' sharing amongst *environmental ethusiasts,* who included the following related experts:

10.2.1–Team Leader, who has appropriate training and wide experience, which includes *livelihoods, project management, food security* etc, to ensure prudent implementation of This Initiative;

10.2.2-Dryland Hydrologist, to ensure the appropriate rainwater ways, cut-offs are improved for appropriate harvesting and utilization, as well as maintained;

10.2.3-Conservatoire of Forests – to coordinate identification of suitable vegetation for environmental recovery;

10.2.4–Social Development Practitioner/*change agent* – to oversee empowerment of affected communities for ownership & sustainability, with focus on ***inclusivity, more so for women, youth and the pwds;***

10.2.5- A Range Grass Practitioneer, *or pasture agronomist* in collaboration with the locals, to assit in the identification and establishment of suitable *trees/grasses/fodders* at every site.

10.2.6–An Environmentalist, *with bias in Climate Change*, to ensure respective concerns are addressed,

10.2.7- Construction Engineer – to lead reconstruction of gullied water ways, broken bridges etc**,** and

10.2.8-A Cartographer*,* to capture a respective map-with legend *(as baseline infor & after)*, depicting target sites, structures in their appropriate locations, *among other relevant expetise.*

11 WORKING BUDGET:

While targeting **1 Catchment** (Site) **in each of the 7**[3] (sample) **dryland Counties,** *depending on the length of the drainage, area, and topography*, **the cost would be about Kshs 15 million, per site, thus a total KES 105 million**[4] **(US $ 800,000); Further***; the actual figure would be determined upon approval & site visit.*

NEVERTHELESS, EITHER OF LAKE BARINGO AND/OR RIVER KWA NGWESU CATCHMENTS IN MAKUENI CATCHMENT WOULD BE A PRIORITY TO CURB EXPANDING DESTRUCTION.

12 CONTACT OF THE INITIATIVE

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[3] ***Baringo, Kilifi, Makueni, Narok, Kajiado, Taita Taveta & Turkana***

[4] ***To construct a 20,000-cubic meter water pan, it would require an additional KES 13 m*** *(whose main costly item is the hiring of an Earthmover; however, it would require less to rehabilitae the nonfunctional ones where they exist)****.***